

Olga Isupova

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Date of Birth	1 st December 1989	Email	ihoho89@gmail.com
Nationality	Russian	Web	profile

Education

- 2014–** *PhD*
2017 [The University of Sheffield](#), UK
Department of [Automatic Control and Systems Engineering](#).
Thesis: *Machine learning methods for behaviour analysis and anomaly detection in video*. Development of methods for autonomous video scene interpretation and abnormal event detection. Methodology included topic modeling, Bayesian nonparametrics, change point detection, approximated and sampling inference schemes, image and video processing.
Supervisor: [Prof. Lyudmila Mihaylova](#), Professor in Signal Processing and Control.
- 2010–** Finished [Yandex School of Data Analysis](#), Russia
2013 Master level program of [Data Analysis](#)
Study on algorithms and data structures, probabilistic methods in decision making theory, machine learning theory, convex theory and optimisation, linguistics and machine translation and others
- 2007–** *Specialist in Mathematics and System Programming* with Honours, degree equals to Masters
2012 [Lomonosov Moscow State University](#), Russia
Faculty of [Computational Mathematics and Cybernetics](#), Department of [Mathematical Methods of Forecasting](#)
Thesis: *Hierarchical topic modeling in document collections by probabilistic latent semantic analysis*. Thesis is devoted to the building a topic hierarchy tree from an unlabelled document collection.
Supervisor: [Prof. Konstantin Vorontsov](#), Lecturer at the Lomonosov Moscow State University

Employment History

- 04.2017 -** *KTP Research Associate*
present [The University of Liverpool](#)
Knowledge transfer partnership [project](#) with [Fletchers Solicitors](#). Development of stand-alone decision making systems on prediction of success of real legal cases based on initial information including structured information, free text clients' stories and evidence records.
- 04.2014 -** *Marie Curie Early Stage Researcher at* [Marie Curie FP7 ITN TRAX project](#)
03.2017 [The University of Sheffield](#)
Research topic is "Knowledge Extraction from data". Development of new algorithms for information extraction from complex, heterogeneous data. Design of a new theory providing computationally feasible ways to uncover the underlying structures in the data. Main applications: anomaly detection and sparse modelling.
- 03.2011 -** *Research Associate*
02.2014 [Croc Inc](#)
Machine learning and data mining research and development: design of data mining algorithms, outliers filtering algorithms; data pre-processing.
Business application functional implementation: design of business processes for data analysis, automation of developed algorithms, customers interviewing.

06.2011 - *Analyst of the Customer Relationship Management department*

09.2011 [Tehnosila](#)

Data analysis in retail operations: sampling customers for target marketing actions; analysis of communication channels, results of marketing actions; statistical and econometric analysis of customers' databases.

Expertise

- **Maths**

Probability theory, Statistics, Stochastic processes
Machine learning, Bayesian nonparametrics, Sparse models
Graphical models, Approximate Inference, MCMC
Mathematical analysis, Convex Optimisation

- **Programming and Miscellaneous**

C/C++, Matlab, R, Python
Databases, SQL

Other research related activity

- **Teaching**

Laboratory assistant and lab tasks designer for the master course "Intelligent and Vision Systems", Department of Automatic Control and Systems Engineering, the University of Sheffield, Spring term 2016. *Feedback from the students is that the labs were the most interesting part of the course*
Help with postgraduate students supervision

- **Talks**

Talk "Modern Video Analytics Systems" at the Sheffield Festival of Science and Engineering as part of British Science week 2016, March 2016;
Talk "Abnormal Behaviour Detection in Video" at the PGR symposium at the University of Sheffield, November 2014

- **Served as chairs and organisers**

A session chair at the 19th International Conference on Information Fusion, July 2016;
The organising committee of the summer school for the TRAX project, Sheffield, June-July 2015;
Athena Swan Committee at the department of ACSE, the University of Sheffield

- **Served as reviewers**

IEEE Transactions of Neural Networks and Learning Systems, Signal Processing Elsevier journal, 2016
American Control Conference, IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI 2016), 29th Conference on Neural Information Processing Systems (NIPS 2016), 2015 American Control Conference

Interests

- **Research**

I really enjoy the process of researching and knowing new things. I want to know novel solutions and methods in my area and related. Therefore I regularly broaden my knowledge and expertise by completing online courses, viewing videolectures and tutorials.

- **Volleyball**

- **Snowboarding, Surfing**

- **Board games**